

# ANGELA NGOZIKA CHUKWUEBUKA

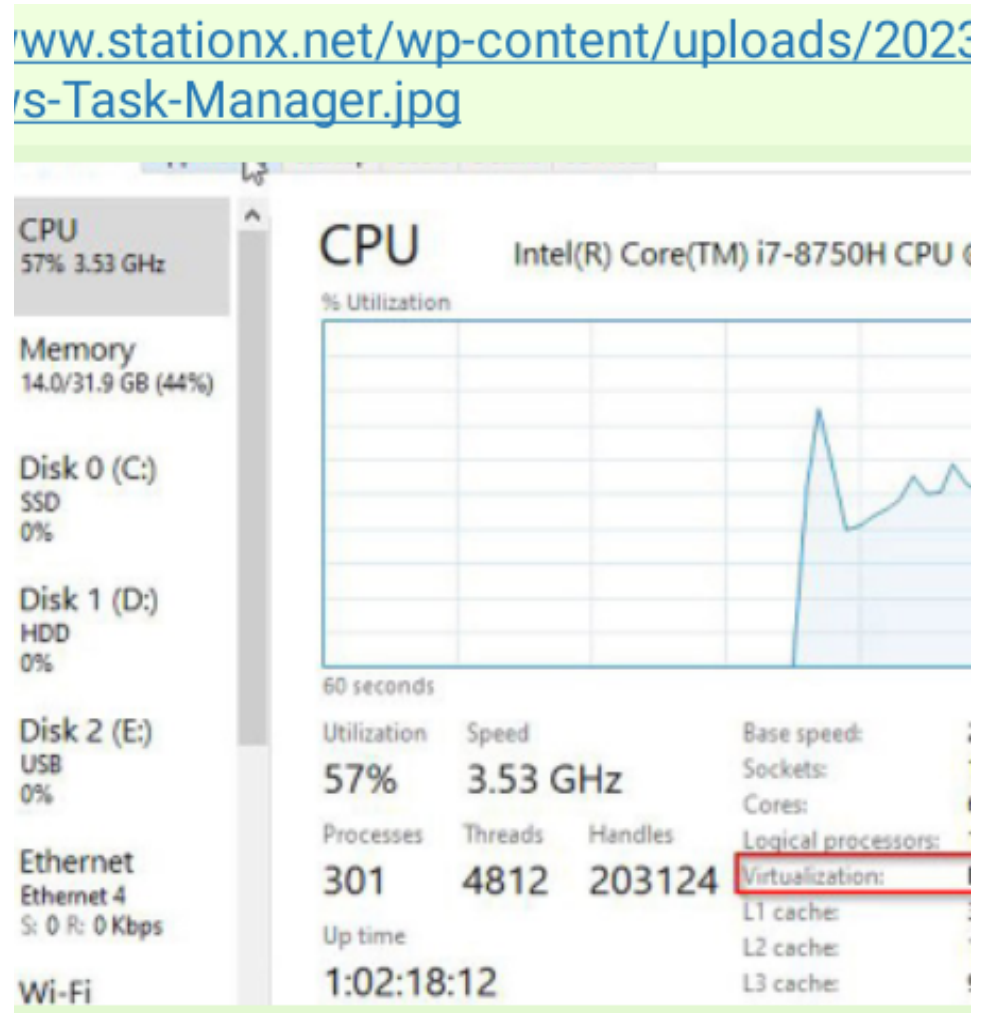
DSA PROJECT



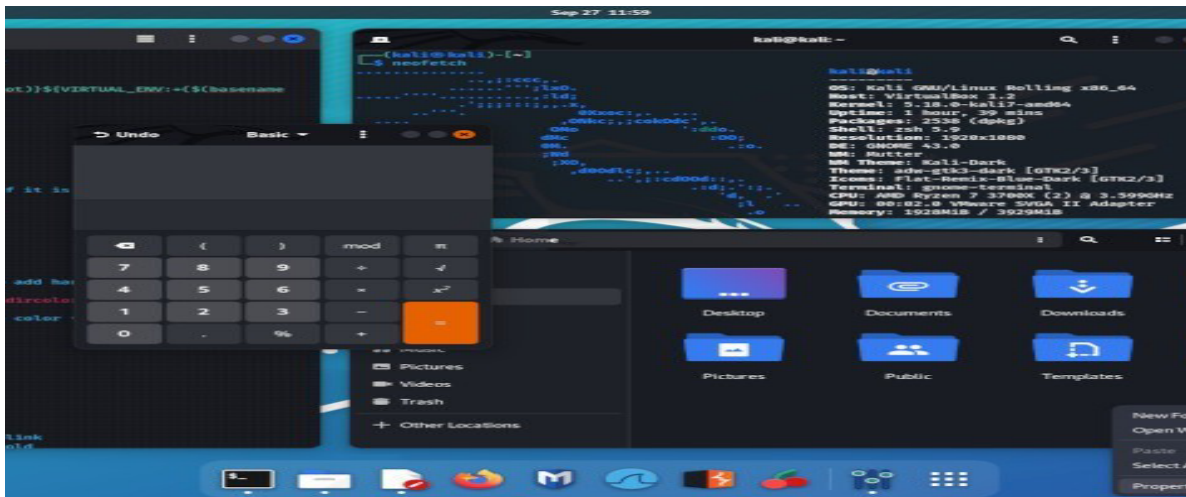
# Project 1: set up a full functional virtual cyber security lab

First of all, I checked my CPU to see if my virtualization is enabled.

On my windows, I clicked on ctrl+shift+esc to open the task manager and then I clicked on the performance tab, then I saw that my virtualization is enabled.



# To set up my lab, I needed the following tools



Hardware: A laptop or desktop computer with sufficient resources. At a minimum, you'll need:

Processor: Dual-core CPU (Intel or AMD) is sufficient, but a quad-core processor is preferable.

RAM: 4GB of RAM will work, but 8GB or more is ideal for running multiple virtual machines.

Storage: At least 20GB for Kali Linux and more if you plan to store files or run additional systems.

Software: The following applications will be needed to set up your virtual lab:

Kali Linux ISO: Download the latest version of Kali Linux from the official website.

Virtualization Software: Options include:

VMware Workstation Player (free for non-commercial use) or VMware Workstation Pro for advanced features.

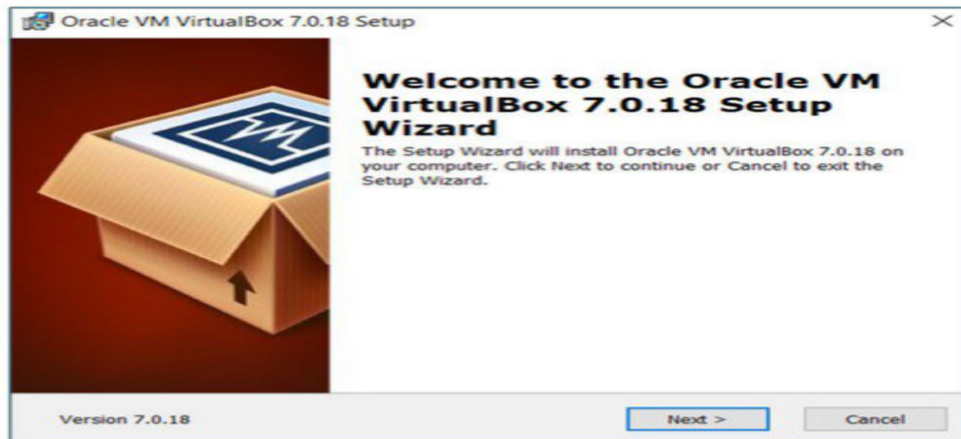
Oracle VirtualBox, a free and open-source virtualization software.

Additional Operating Systems: If you wish to set up vulnerable machines, consider downloading pre-made virtual machine images, such as Metasploitable or OWASP's BWA.

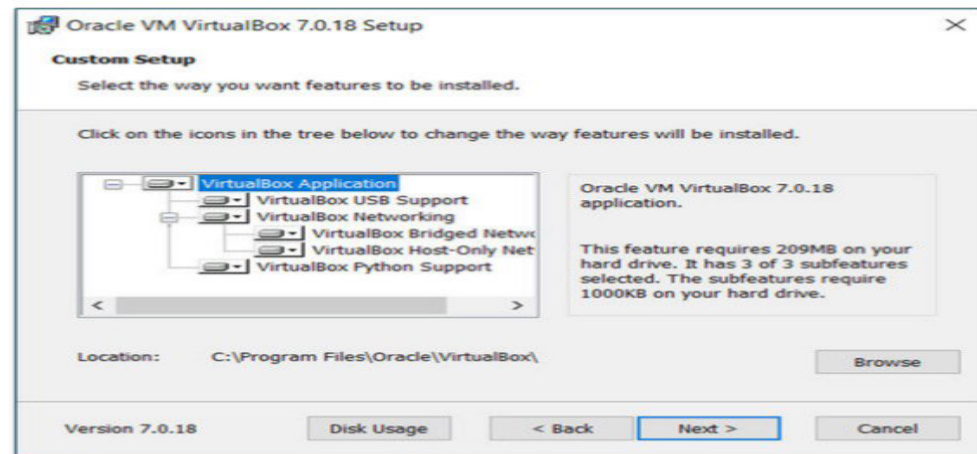
Networking Equipment: A basic wireless router and switches are enough for a simple setup. You can also create isolated networks using virtual adapters within your virtualization software.

First step is to download Oracle virtual box from <https://www.virtualbox.org/wiki/Downloads>

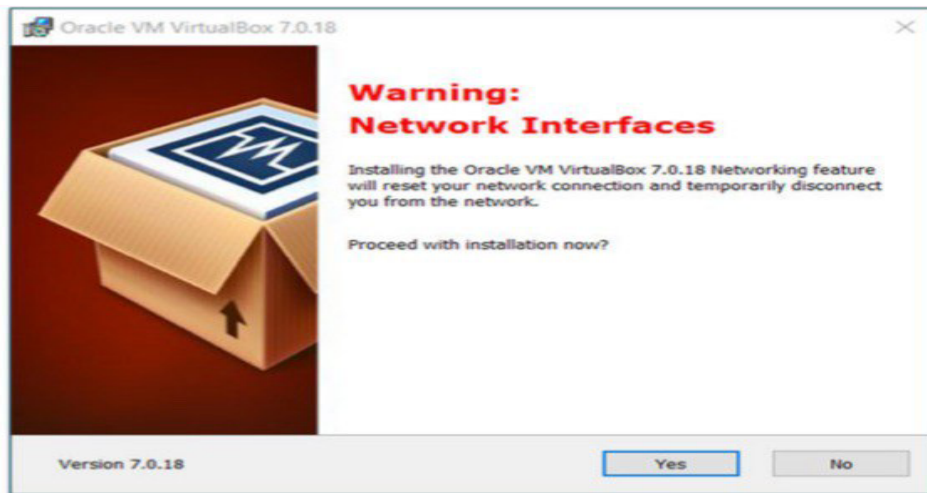
Go to your downloads folder and double click the VirtualBox executable to install it.



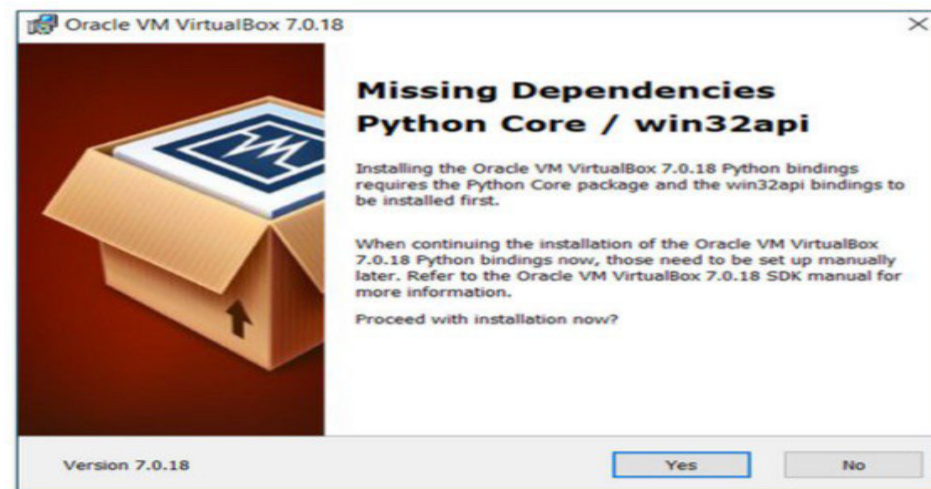
Click **Next** 



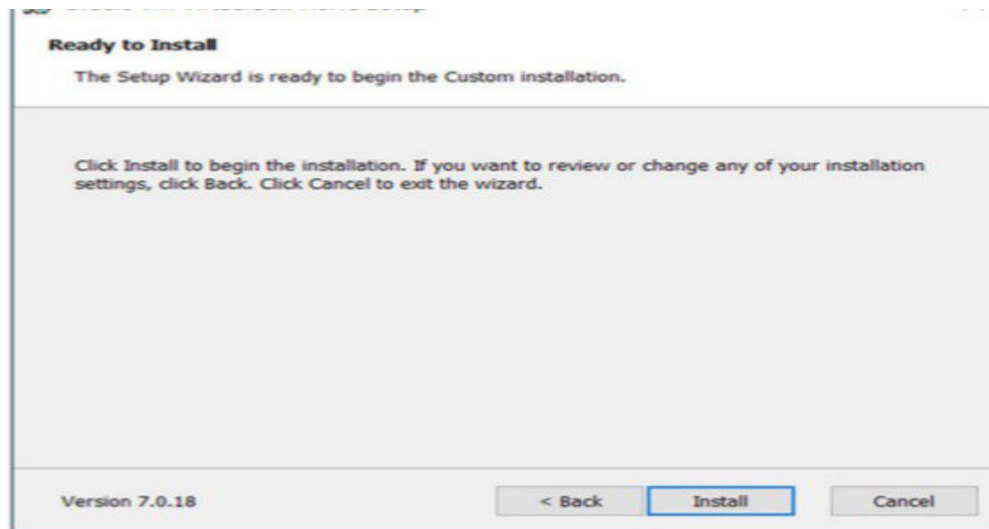
Click **Next** 



Click **YES** 



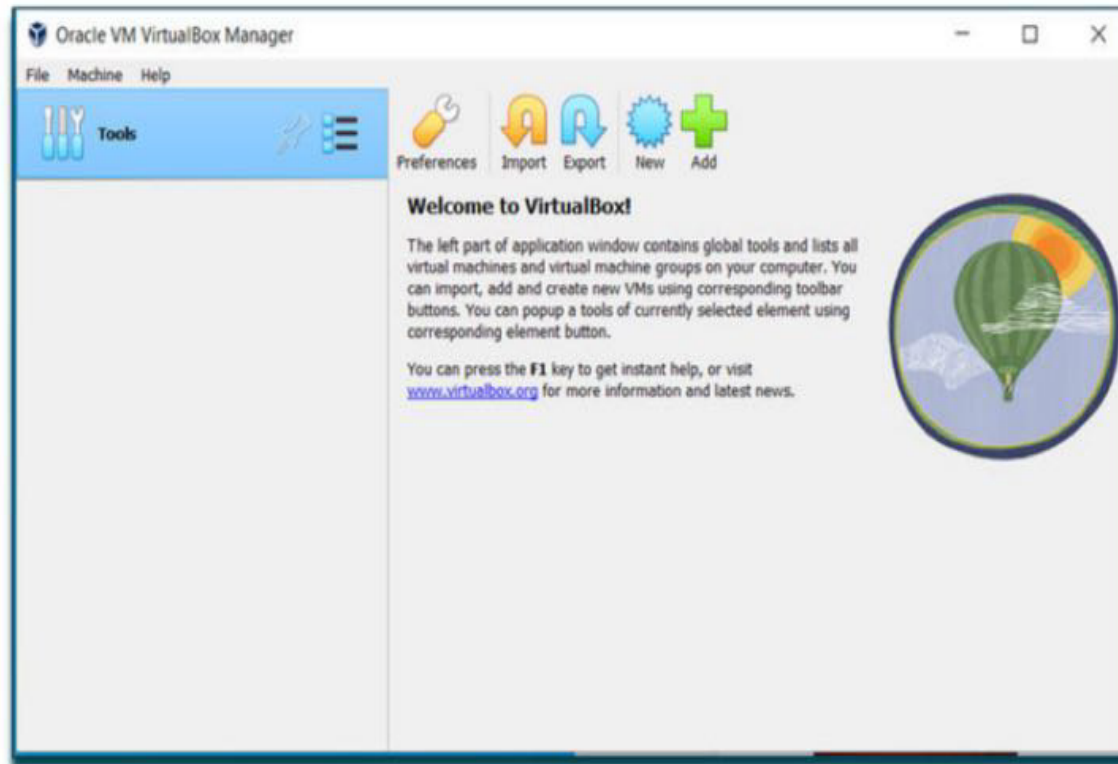
Click **YES** 



Click **Install** ➡



Click **Finish** ➡

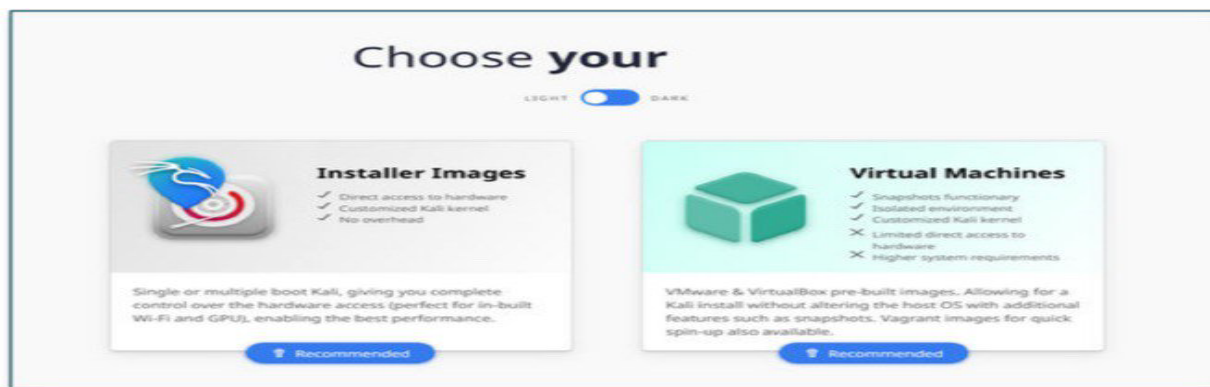


**Congratulations, you have now successfully installed VirtualBox.**

From here we can create both virtual machines to test from (typically Kali) and virtual machines to attack (anything we choose).

Now to create a testing machine, we have to download Kali from <https://www.kali.org/get-kali/#kali-platforms>, click on virtual machine and click on 64 bit.

You will have two options when it comes to downloading Kali – either to download a pre-built virtual machine, or the image installer (a .iso file)



The key difference is that downloading the ISO file will let you install Kali Linux from scratch, giving you more control over the setup, while the pre-built VM is a ready-to-use version that you can import directly into your virtualisation software, saving time and effort.

First, let's start with using a **prebuilt machine**.

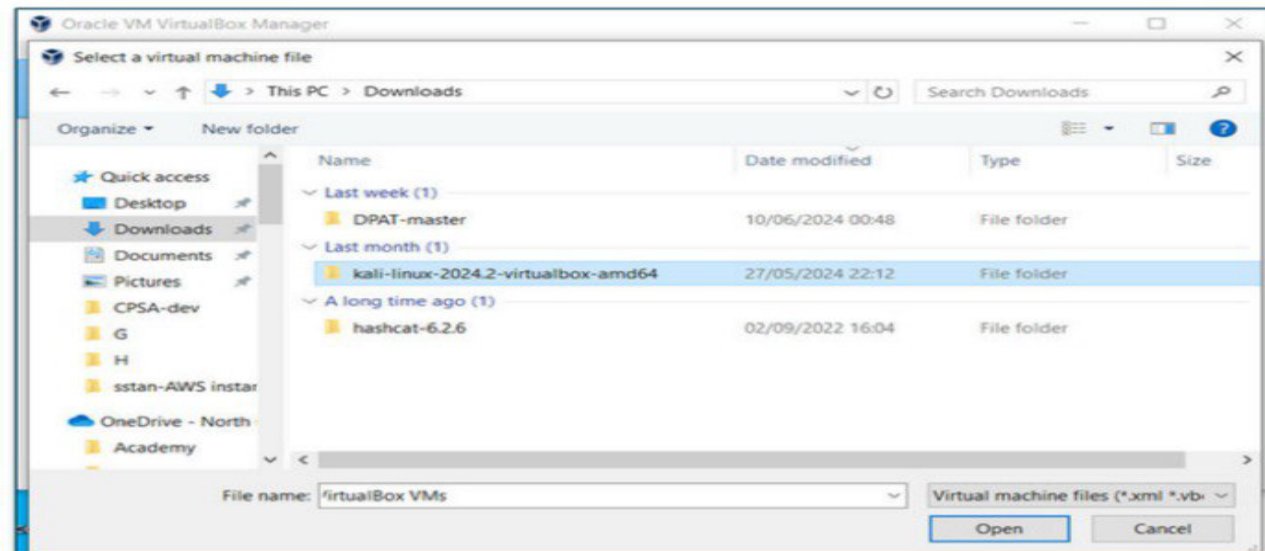


Select the software you use (we are going to assume VirtualBox). This will give you a zipped file.

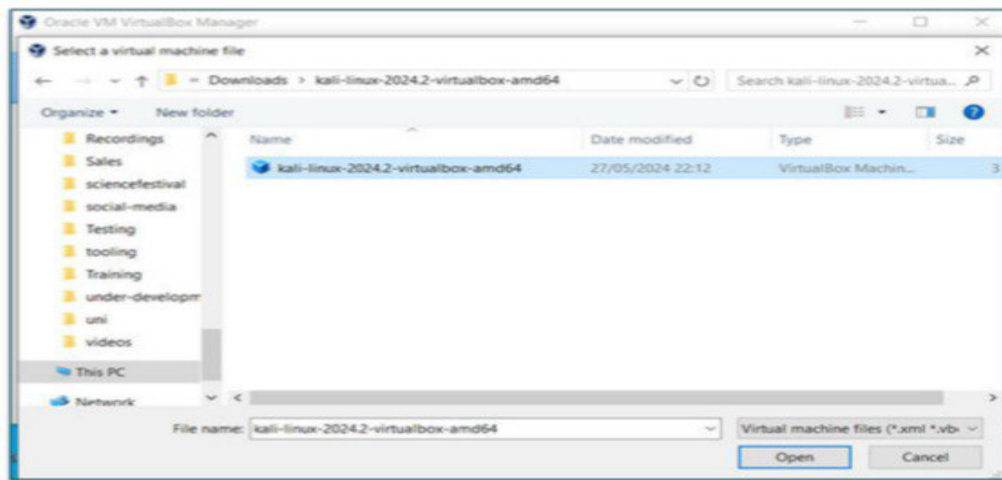
To access this virtual machine, simply **Add** it to VirtualBox.



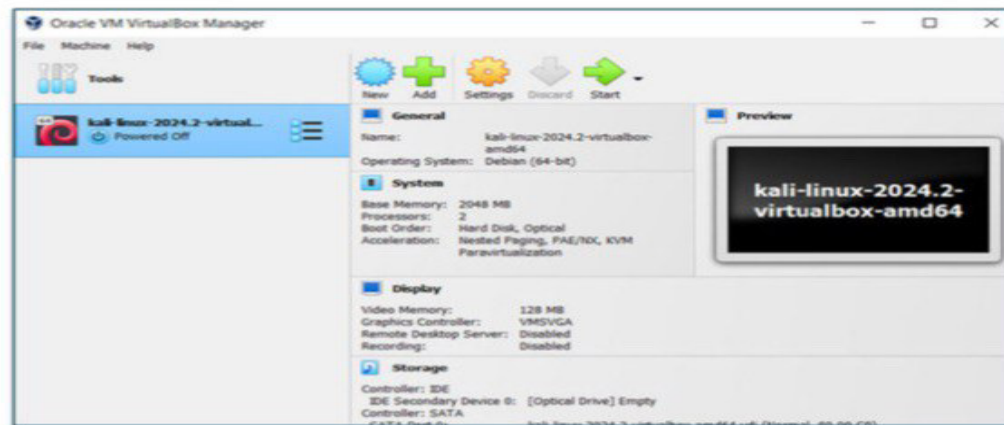
Click **Add** ➡



Select the **Kali Linux** folder ➡



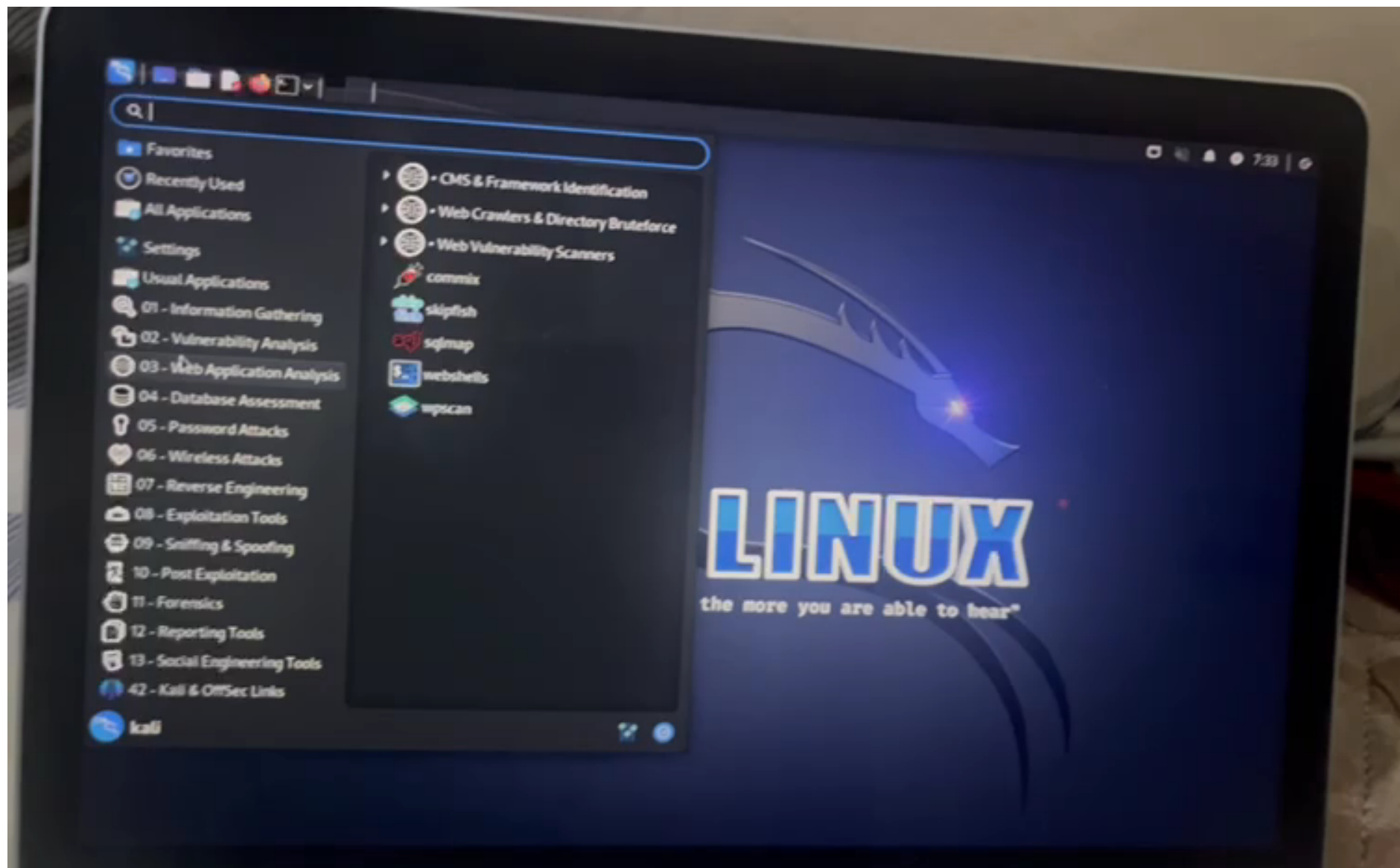
And choose the **Kali VirtualBox** file 



Your Kali machine now appears in your VirtualBox interface.

Pressing start will boot up the machine. It will use the default credentials of **kali:kali**

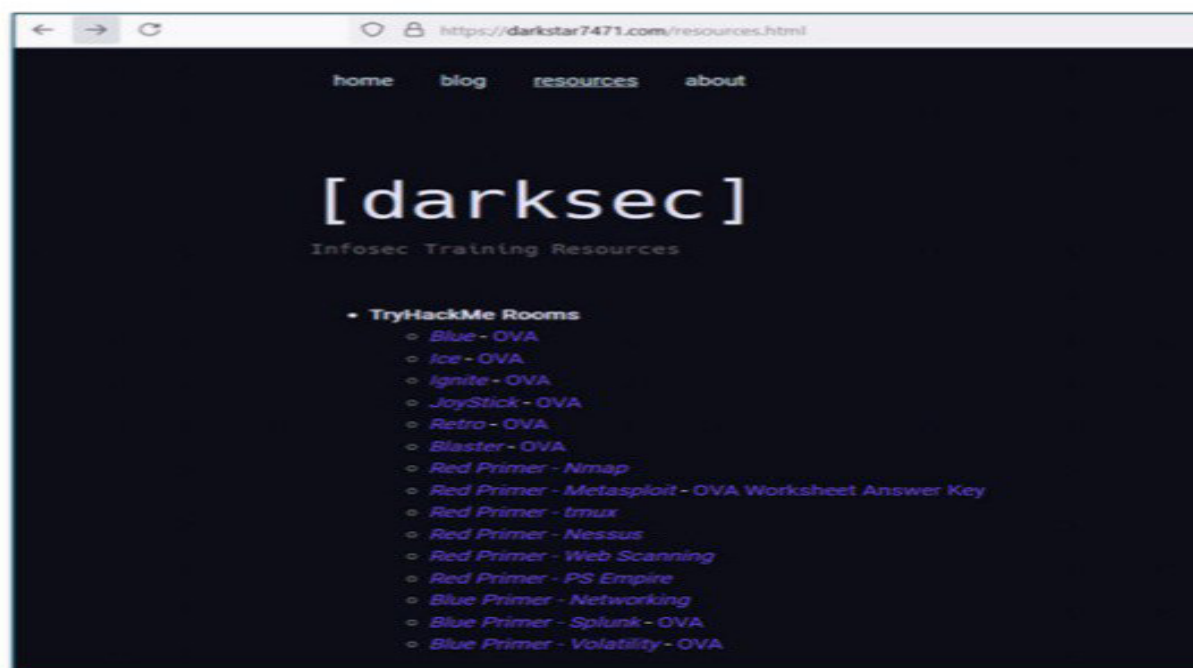
You will now be able to use your Kali virtual machine.



Here is my live Kali, you can see all the application On my sytem .

have a Windows machine to test against, we will use the machine **Blue** from **Hack The Box**. This is a commonly used vulnerable Windows machine, again with plenty of write ups available across the internet in case you get stuck.

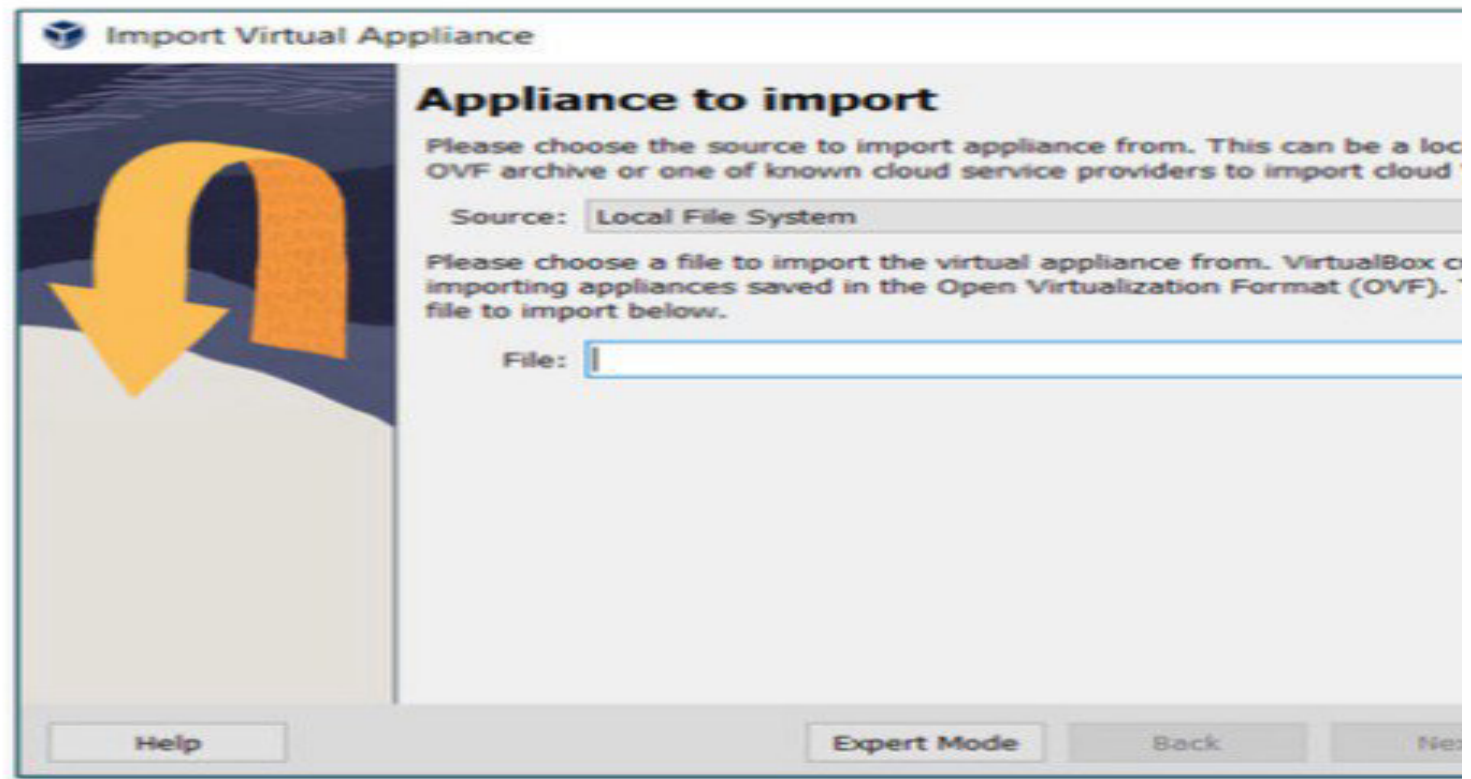
to get started, download **Blue** from <https://darkstar7471.com/resources.html> choosing the **OVA** link next to Blue.



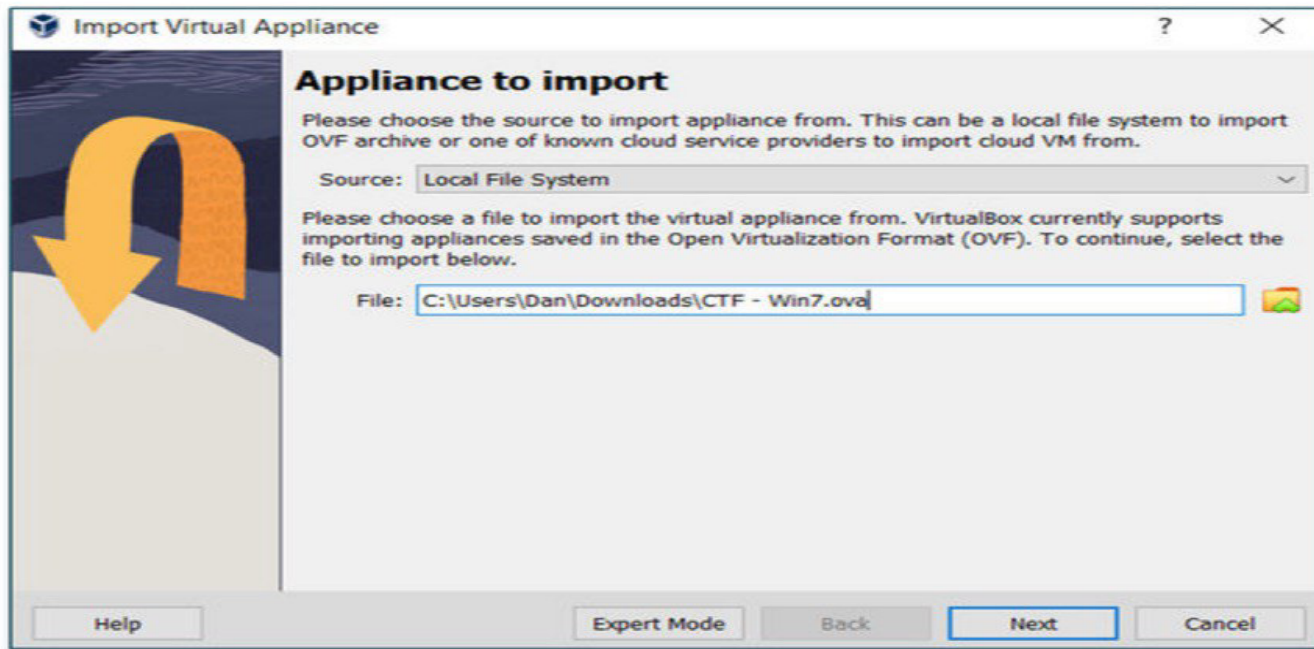
this will download a zip file that contains the prebuilt VM called **CTF - Win7**.

We can now import this file into VirtualBox as an appliance, by navigating to **File, Import Appliance**.

Or you can download windows 10 from <https://www.microsoft.com/en-us/software-download/windows10>, and import it into the Virtual box.



simply select the **CTF – Win 7 OVA** file and hit **Next** 



Then **Finish** 

Congratulations! I have set up my very own lab environment with virtual box.

Virtual lab provides a safe space to explore, test and refine cyber security skills without the risk of causing harm to physical system/devices.